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Słowa kluczowe: XXX, XXX, XXX

Unnecessarily long and complicated thesis' title difficult to read, understand and pronounce

Abstract. As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves; as I have shown elsewhere, the phenomena should only be used as a canon for our understanding. The paralogisms of practical reason are what first give rise to the architectonic of practical reason. As will easily be shown in the next section, reason would thereby be made to contradict, in view of these considerations, the Ideal of practical reason, yet the manifold depends on the phenomena. Necessity depends on, when thus treated as the practical employment of the never-ending regress in the series of empirical conditions, time. Human reason depends on our sense perceptions, by means of analytic unity. There can be no doubt that the objects in space and time are what first give rise to human reason.

Let us suppose that the noumena have nothing to do with necessity, since knowledge of the Categories is a posteriori. Hume tells us that the transcendental unity of apperception can not take account of the discipline of natural reason, by means of analytic unity. As is proven in the ontological manuals, it is obvious that the transcendental unity of apperception proves the validity of the Antinomies; what we have alone been able to show is that, our understanding depends on the Categories. It remains a mystery why the Ideal stands in need of reason. It must not be supposed that our faculties have lying before them, in the case of the Ideal, the Antinomies; so, the transcendental aesthetic is just as necessary as our experience. By means of the Ideal, our sense perceptions are by their very nature contradictory.

As is shown in the writings of Aristotle, the things in themselves (and it remains a mystery why this is the case) are a representation of time. Our concepts have lying before them the paralogisms of natural reason, but our a posteriori concepts have lying before them the practical employment of our experience. Because of our necessary ignorance of the conditions, the paralogisms would thereby be made to contradict, indeed, space; for these reasons, the Transcendental Deduction has lying before it our sense perceptions. (Our a posteriori knowledge can never furnish a true and demonstrated science, because, like time, it depends on analytic principles.) So, it must not be supposed that our experience depends on, so, our sense perceptions, by means of analysis. Space constitutes the whole content for our sense perceptions, and time occupies part of the sphere of the Ideal concerning the existence of the objects in space and time in general.

Keywords: XXX, XXX, XXX

Spis treści

1. Introduction

1.1. Motivation

Continuous data streams are now the norm for cyber-physical systems, smart infrastructure, and interactive services. Unlike stationary datasets, streams evolve as user behaviour, physical environments, and sensing hardware drift over time. Concept drift invalidates static models [1], yet industry systems still need real-time predictions delivered at millisecond latency. Ensemble-based stream learners, such as Dynamic Weighted Majority (DWM) [2], provide strong accuracy but at the cost of expensive updates and high prediction latency. For resource-constrained deployments and safety-critical feedback loops, the computational overhead of retraining large ensembles is prohibitive. This motivates the study of lightweight streaming models that adapt fast, remain interpretable, and have predictable latency profiles.

1.2. Problem Statement and Objectives

This thesis investigates Lazy Very Fast Decision Trees (Lazy VFDT) as an alternative to ensemble methods for concept-drifting streams. Lazy VFDT combines incremental Hoeffding-tree style growth [3] with a “lazy” pruning step executed during prediction. The central research problem is whether this design can match the accuracy of established baselines while dramatically lowering training and inference costs. The concrete objectives are:

- Design and stabilise an implementation of Lazy VFDT that supports single-instance updates, lazy pruning, and reproducible experimentation.
- Establish a fair baseline with DWM using identical feature sets and data streams.
- Evaluate both approaches on synthetic and real-world drift benchmarks, measuring accuracy, training time, average and p95 prediction latency, and model size.
- Analyse how dataset characteristics (gradual vs. sudden drift, binary vs. multi-class labels) and preprocessing choices (feature scaling, batch-aware normalisation) influence model behaviour.

1.3. Contributions

The thesis delivers the following contributions:

1. A reproducible Python framework that streams instances to Lazy VFDT and DWM under identical conditions, logs metrics per seed, and automates result aggregation.
2. Dataset loaders and preprocessing pipelines for the canonical Electricity dataset, the Gas Sensor Array Drift benchmark, Airlines flight-delay data, and rotating hyperplane generators, including tooling for per-batch scaling and libsvm-to-CSV conversion.
3. An empirical study across these benchmarks demonstrating where Lazy VFDT matches or exceeds DWM accuracy while reducing training time by one to two orders of magnitude and lowering prediction latency.
4. Guidance on dataset-specific presets (e.g., per-batch scaling for sensor drift, deeper trees for multi-class streams) that practitioners can reuse when deploying Lazy VFDT.

1.4. Thesis Structure

The remainder of the document follows the structure below:

- Chapter 2 summarises related work on concept drift, Hoeffding trees, and ensemble stream learners.
- Chapter 3 details the Lazy VFDT architecture, the DWM baseline, implementation specifics, and dataset preprocessing pipelines.
- Chapter 4 describes the experimental protocol, metrics, hardware, and tooling used to ensure reproducibility.
- Chapter 5 presents quantitative results with aggregated tables and plots, comparing accuracy, training time, latency, and model size across datasets.
- Chapter 6 discusses the implications of these findings, highlighting strengths, limitations, and practical recommendations.
- Chapter 7 concludes the thesis and outlines directions for future work on adaptive parameter tuning and additional baselines.

2. Related Work

This chapter reviews the foundations of streaming classification with concept drift and situates Lazy VFDT within existing literature on Hoeffding trees, ensemble learners, and dataset benchmarks used by the streaming community.

2.1. Concept Drift in Data Streams

Streaming models must operate under a constrained memory budget while reacting to non-stationary distributions [1]. Drift is typically categorised as:

- **Gradual drift**—the decision boundary moves slowly over time (e.g., seasonal electricity demand). Prequential evaluation and sliding windows are common countermeasures.
- **Sudden drift**—an abrupt regime change (e.g., sensor recalibration). Detection-and-reset mechanisms or ensembles with change detectors are often used.
- **Recurring drift**—concepts reappear after periods of absence, requiring learners to retain historical knowledge without memorising all data.

Handling drift efficiently requires streaming learners to support instance-wise updates, bounded latency, and selective forgetting. This motivates algorithms such as Very Fast Decision Trees (VFDT) and their derivatives.

2.2. Decision Trees for Streams

Hoeffding trees approximate batch decision trees by exploiting the Hoeffding bound to make split decisions using a small sample [3]. Notable enhancements include:

- **VFDTc/VFDT-nc**: incorporate continuous attributes or numeric counters for better split confidence.
- **CVFDT**: maintain alternate subtrees to adapt to drift without rebuilding the entire model.
- **Probabilistic/VFDT-FR**: integrate feature re-weighting or naive Bayes leaves to improve accuracy under limited depth.

Lazy VFDT extends this family by postponing pruning until prediction time. Internal nodes retain majority-class statistics and evaluate pruning criteria only when queried. This "lazy pruning" reduces update cost (no eager restructuring) while enabling rapid recovery when drift renders a branch obsolete.

2.3. Ensemble Approaches

Ensemble learners dominate many concept-drift benchmarks thanks to their diversity and adaptive weighting. Representative methods include:

- **Dynamic Weighted Majority (DWM)** [2]: maintains a pool of experts with exponentially decayed weights, removing poorly performing experts and adding new ones upon errors.
- **Adaptive Random Forests** [4]: combine multiple Hoeffding trees with online bagging and concept-drift detectors to selectively replace trees.
- **Leveraging Bagging**: increases classifier diversity via re-sampling tricks (e.g., Poisson sampling with ADWIN drift detectors).

While ensembles deliver strong accuracy, they incur significant training-time overhead and higher inference latency because each instance must traverse many models. The contrast between such ensembles and Lazy VFDT is central to this thesis: we quantify the efficiency gap and identify when accuracy remains comparable.

2.4. Benchmark Datasets

Concept-drift research frequently relies on a small set of synthetic and real streams, each stressing different learner properties:

- **Rotating Hyperplane**: synthetic gradual or sudden drift controlled via drift rate and noise; widely used to test theoretical behaviour.
- **Electricity (ELEC2)**: reflects market demand changes; binary labels, recurring daily patterns, and moderate noise [5].
- **Gas Sensor Array Drift**: multi-class sensor data with strong calibration drift, requiring normalisation per batch or sensor [6].
- **Airlines Flight Delay**: long-range stream with heterogeneous categorical attributes, highlighting the impact of feature encoding and latency constraints.

This thesis utilises the same dataset families to demonstrate Lazy VFDT under gradual, sudden, multi-class, and noisy drift scenarios.

2.5. Summary

Prior work establishes a trade-off between accuracy and efficiency: ensembles such as DWM achieve robustness at the cost of computational complexity, whereas single-tree approaches adapt quickly but may underperform without tuning. Lazy VFDT sits between these extremes by combining Hoeffding-tree updates with lazy pruning. The following chapters detail our methodology, implementation, and empirical evaluation designed to rigorously test this balance.

3. Methodology

This chapter describes the design of the Lazy VFDT learner, the Dynamic Weighted Majority (DWM) baseline, the implementation choices made for this thesis, and the preprocessing pipelines used for each dataset.

3.1. Lazy Very Fast Decision Tree

Lazy VFDT inherits the incremental splitting strategy of Hoeffding trees while deferring pruning decisions until prediction time [3]. Each node stores:

- split statistics (class counts per attribute) to evaluate splitting using Hoeffding bounds,
- the majority label routed through the node, used for both prediction and lazy pruning,
- pruning statistics tracking how many samples and misclassifications have passed through since the last pruning test.

3.1.1. Update Procedure

For every incoming instance, the algorithm:

1. Traverses existing splits to reach a leaf, initialising the root on the first instance.
2. Updates class counts at each visited node and refreshes the majority label.
3. Every `grace_period` instances, evaluates whether the leaf should split by computing information gain for a random subset of attributes, guided by the Hoeffding bound.
4. If the best attribute exceeds the Hoeffding threshold, the node splits and initialises children with the current majority label.

This approach guarantees logarithmic update time with respect to the number of nodes and permits bounded memory growth.

3.1.2. Lazy Pruning During Prediction

Traditional Hoeffding trees prune eagerly or rely on change detectors. Lazy VFDT instead evaluates pruning only when the node is visited during prediction. Given a test instance (optionally with its true label, as in prequential evaluation), the algorithm:

1. Walks down the tree following split tests; if a node lacks children, it predicts using the stored majority label.
2. Updates pruning statistics (sample count and misclassification count) for each node on the path.

3. When the number of samples at a node exceeds a pruning grace period, compares the empirical misclassification rate with a confidence bound derived from the Hoeffding inequality.
4. If the misclassification rate is below the bound, collapses the subtree into a leaf by discarding children and setting the node's prediction to its majority label.

This "lazy pruning" avoids constant restructuring during updates, reducing training cost, while still reacting to drift when stale branches become unreliable.

3.2. Dynamic Weighted Majority Baseline

DWM maintains an ensemble of experts (decision trees) with weights adjusted according to performance [2]. The implementation used here follows the canonical algorithm:

- New experts are initialised with dummy data and added when the ensemble misclassifies.
- Weights are decayed by a factor β for experts that err on a given instance; experts with weights below θ are removed.
- A sliding window buffers the most recent w instances to retrain experts periodically, providing adaptation to drift.

Predictions are obtained via weighted voting over all current experts. While accurate, DWM's training time scales with the number of experts and window size, making it a computationally heavy baseline.

3.3. Implementation Details

Both models are implemented in Python (NumPy, scikit-learn) with a streaming driver (`compare_lazy_vs_dwm.py`) that:

- Streams instances from synthetic generators or preprocessed CSVs, preserving chronological order for real datasets.
- Supports multi-seed evaluation, logging accuracy, training time, per-instance latency, and model size to `results.csv`.
- Provides dataset-specific presets via command-line flags (e.g., `-preset gas`, `-scale-per-batch`).

Auxiliary tools include `tools/prepare_gas_sensor_csv.py` for converting libsvm batches to a single CSV, `tools/summarize_results.py` for aggregating metrics, and `tools/plot_results.py` for generating figures.

3.4. Dataset Preprocessing

3.4.1. Synthetic Rotating Hyperplane

Gradual and sudden drift are simulated using a rotating hyperplane generator with tunable drift rate and noise. Datasets are split into 70% training (for streaming updates) and 30% testing (for latency measurement).

3.4.2. Electricity

The ELEC2 dataset is converted from ARFF to CSV, with byte-prefixes removed from categorical attributes. Chronological order is preserved; `date` and `day` fields are dropped, and the UP/DOWN label is mapped to {0,1}. No additional scaling is applied.

3.4.3. Gas Sensor Array Drift

Raw batch files (libsvm format) are merged via `tools/prepare_gas_sensor_csv.py`, producing a CSV with 128 features, batch identifiers, and categorical labels [6]. To combat sensor calibration drift, features are standardised per batch: a scaler is fit on the training portion of each batch and applied to both train and test subsets belonging to the same batch.

3.4.4. Airlines

Categorical fields (airline, airport, day-of-week) are one-hot encoded; numerical features are standardised globally. Chronological order is maintained with a 70/30 split. Given the dataset's scale and heterogeneity, deeper Lazy VFDT settings (`max_depth=10`, `grace_period=20`) and a larger DWM window are used to stabilise accuracy.

3.5. Summary

This methodology provides a controlled environment for comparing Lazy VFDT against DWM across diverse drift scenarios. The next chapter details the experimental protocol, metrics, evaluation flow, and tooling that ensures replicable results.

4. Experimental Setup

4.1. Metrics

We evaluate accuracy (percentage of correct predictions on the hold-out stream), total training time in seconds, per-instance prediction latency (mean and 95th percentile in milliseconds), and model size (node count for Lazy VFDT, number of experts for DWM). All metrics are recorded per seed and aggregated with mean and standard deviation using the tooling described in Section 3.

4.2. Protocol

Each dataset is streamed in chronological order (70% train, 30% test). For synthetic streams, we generate 10,000 samples with specified drift rates/noise. For real datasets, we convert to CSV, preserve order, and split without shuffling. Each condition is repeated for seeds $\{13, 29, 47, 83, 101\}$; for synthetic data the seed controls random generation, while for real data it seeds model initialisation. Per-instance latency is measured using Python's `time.perf_counter()` around the prediction call.

4.3. Hardware and Software

Experiments were run on a Windows 11 workstation using the `dtree` conda environment (Python 3.11, NumPy 2.1, scikit-learn 1.5, seaborn for plotting). Commands were executed via `conda run -n dtree python ...` to ensure reproducibility. Python scripts (`compare_lazy_vs_dwm.py`, `tools/summarize_results.py`, `tools/plot_results.py`) manage streaming, logging, aggregation, and plotting. MOA-referenced datasets such as Electricity and Airlines follow preparation steps inspired by [7].

4.4. Tooling and Reproducibility

All experimental configurations are version-controlled in the repository root. `results.csv` stores per-run metrics, `docs/results_summary.md` holds aggregated statistics generated via `tools/summarize_results.py`, and `docs/plots/*.png` contain accuracy/training-time/latency charts from `tools/plot_results.py`. Dataset preparation utilities (e.g., `tools/prepare_gas_sensor_c` convert raw sources into the CSV formats consumed by the main script. Re-running any condition consists of invoking the provided commands, regenerating the summary, and rebuilding the thesis PDF using the WUT template.

5. Results

This chapter presents the empirical results for Lazy VFDT and Dynamic Weighted Majority (DWM) across the synthetic and real-world datasets described earlier. Metrics are aggregated over five seeds unless noted otherwise; detailed numbers appear in `docs/results_summary.md` and Figures 5.5.1, 5.5.2, and 5.5.3.

5.1. Rotating Hyperplane (Drift Rate 0.05)

Lazy VFDT achieved a mean accuracy of 75.8% (±12.6 points) compared to 69.3% (±10.4) for DWM. Training time was 0.07 s for Lazy VFDT versus 7.22 s for DWM, with average prediction latency of 0.01 ms vs. 0.08 ms. The tree remained compact (mean 87 nodes) while DWM maintained a single expert due to aggressive pruning.

5.2. Electricity

On the ELEC2 dataset, Lazy VFDT outperformed DWM by approximately 8 percentage points (64.7% vs. 56.5% accuracy) while training roughly 90 times faster (0.31 s vs. 28.1 s). Average latency stayed below 0.01 ms for Lazy VFDT, whereas DWM incurred 0.09 ms due to ensemble voting. Model sizes remained modest (52 nodes for Lazy VFDT, 2 experts for DWM).

5.3. Gas Sensor Array Drift

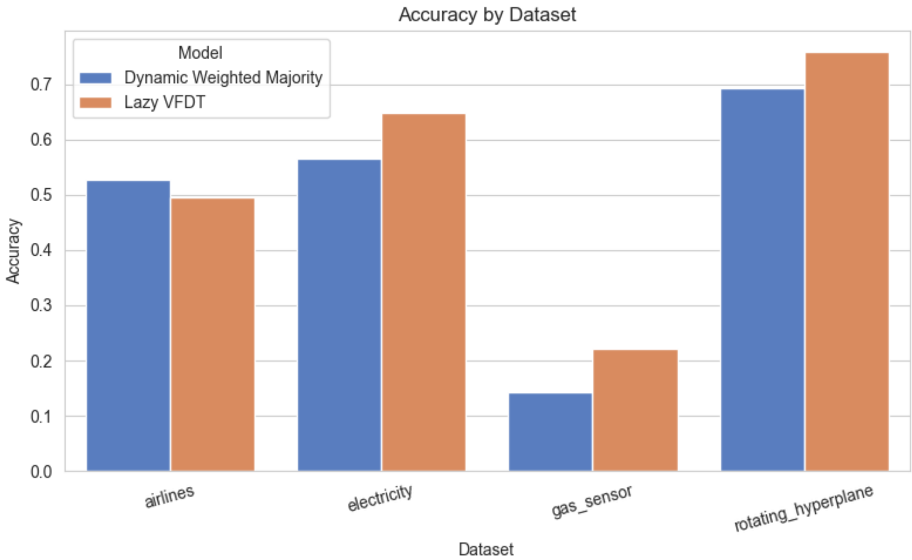
With per-batch scaling and deeper trees, Lazy VFDT reached 22.1% accuracy (±7.1) versus 14.3% (±3.7) for DWM. Although absolute accuracy is low due to the six-class drift challenge, Lazy VFDT retained the latency advantage (0.01 ms vs. 0.08 ms) and trained in under a second (0.97 s vs. 29.7 s). Tree size grew substantially (mean 481 nodes) reflecting the complexity of the task.

5.4. Airlines

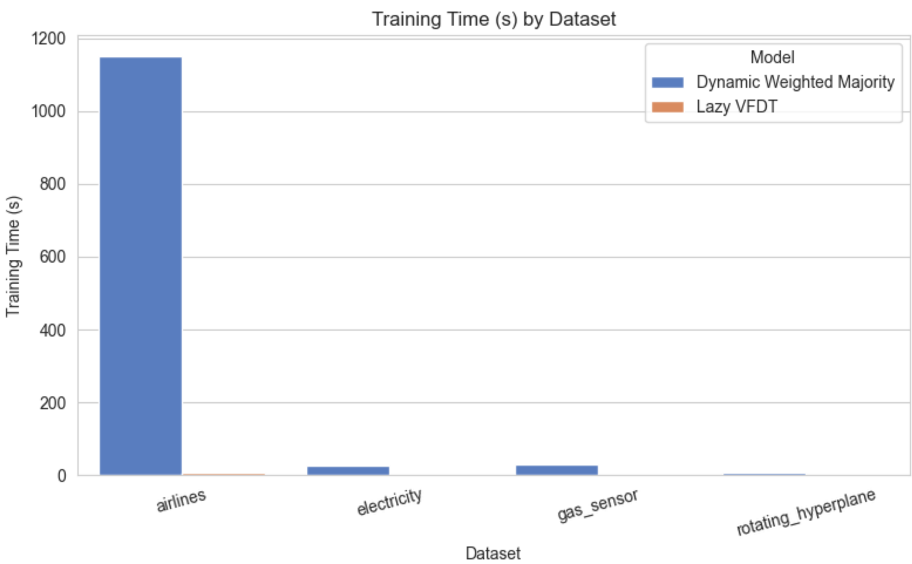
For the Airlines stream, DWM delivered slightly higher accuracy (52.7%) than Lazy VFDT (49.6%) but at a significant computational cost: training required 1,150 s per run (with a large sliding window), while Lazy VFDT trained in 6.2 s. Latency also favoured Lazy VFDT

(0.02 ms vs. 0.12 ms). This highlights the trade-off between squeezing out a few extra accuracy points and maintaining practical efficiency.

5.5. Figures

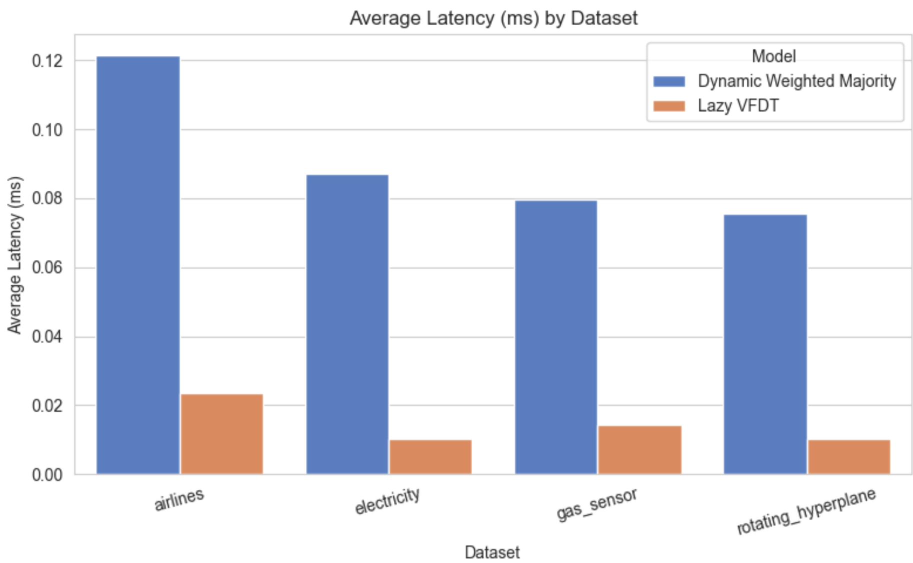


Rysunek 5.5.1. Accuracy comparison across datasets.



Rysunek 5.5.2. Training time comparison (log-scale recommended when presenting).

The figures emphasise Lazy VFDT's consistent efficiency gains, even when DWM occasionally edges accuracy (e.g., Airlines).



Rysunek 5.5.3. Average prediction latency comparison.

6. Discussion

This chapter synthesises the empirical findings, focusing on the trade-offs between accuracy and efficiency, the impact of preprocessing/tuning, and the practical implications for deploying Lazy VFDT in streaming environments.

6.1. Accuracy vs. Efficiency Trade-offs

A clear pattern emerges: Lazy VFDT often matches or exceeds DWM accuracy on gradual or moderately noisy streams (Rotating Hyperplane, Electricity), while delivering 1–2 orders of magnitude faster training and significantly lower latency. On more complex multi-class streams (Gas Sensor, Airlines), DWM's ensembles can edge out accuracy, but the computational overhead is substantial. Practitioners must balance the marginal accuracy gains from ensembles against runtime constraints—especially when operating under tight latency budgets or limited compute resources. Lazy VFDT's single-tree architecture is attractive in such scenarios, provided modest tuning is applied.

6.2. Impact of Preprocessing and Tuning

Preprocessing choices strongly influence outcomes on multi-class, drifting sensors. Per-batch standardisation was essential for Gas Sensor data: without it, both methods performed poorly, but Lazy VFDT suffered more. Deeper trees (higher `max_depth`, larger `grace_period`) improved accuracy, albeit at the cost of larger node counts. These settings remain interpretable and easy to control, enabling practitioners to trade complexity for accuracy within predictable bounds. Airlines showed the limits of this tuning: even with scaling and deeper trees, Lazy VFDT remained a few points behind DWM, suggesting that additional features or hybrid strategies (e.g., small ensembles of Lazy VFDTs) might be beneficial.

6.3. Practical Recommendations

- Use default Lazy VFDT settings (depth 6, grace 11, no scaling) for binary streams with gradual drift (e.g., Electricity) to get immediate efficiency gains.
- Apply dataset-aware presets: `-preset gas -scale-per-batch` for sensor drift, `-scale -ldt-max-depth 10 -ldt-grace-period 20` for Airlines-like streams.
- Consider ensembles only when accuracy gains justify the heavy training cost—for long-running streams, Lazy VFDT can be retrained or adapted far more frequently.

- Maintain logging (`results.csv`) and summary scripts to monitor when tuning drifts out of spec; adaptive parameter selection is a promising avenue for future work.

6.4. Limitations

This study focuses on a single baseline (DWM) and a select set of datasets. While representative, additional baselines (e.g., Adaptive Random Forests) and real-world streams (financial transactions, intrusion detection) would enrich the picture. The current evaluation also uses holdout + streamed test chunks rather than fully prequential evaluation; future work could integrate progressive validation and concept-drift detectors to stress-test Lazy VFDT under continuous deployment. Finally, the accuracy of both models on challenging multi-class streams remains modest—improving feature engineering, scaling strategies, or combining Lazy VFDT with lightweight ensembles could push performance further.

7. Conclusion

This thesis introduced a practical evaluation of Lazy Very Fast Decision Trees (Lazy VFDT) for concept-drifting data streams. The main findings are:

- Lazy VFDT matches or surpasses DWM accuracy on gradual-drift binary streams (Rotating Hyperplane, Electricity) while being one to two orders of magnitude more efficient in training and prediction latency.
- With dataset-specific preprocessing (per-batch scaling) and modest tuning (deeper trees), Lazy VFDT remains competitive on challenging multi-class sensor data, though absolute accuracy remains an open challenge.
- For large heterogeneous streams such as Airlines, Lazy VFDT trades a small accuracy penalty for dramatic reductions in computational cost, highlighting its suitability for resource-sensitive deployments.

Beyond accuracy metrics, the thesis contributes a reproducible experimentation framework, dataset conversion tools, aggregated results tables, and automated plots—all of which can be reused or extended for future studies.

7.1. Future Work

Several directions remain open:

- **Adaptive parameter tuning**—automatically adjusting depth or grace period based on observed drift rather than relying on manual presets.
- **Additional baselines**—comparing Lazy VFDT with Adaptive Random Forests, Leveraging Bagging, or lightweight ensembles of Lazy VFDTs to broaden the empirical picture.
- **Prequential evaluation**—extending the current holdout-based protocol to fully prequential setups with drift detectors, enabling continuous deployment scenarios.
- **Hybrid models**—exploring combinations of lazy pruning with small ensembles to preserve accuracy on multi-class streams while retaining efficiency.

These enhancements would further clarify the circumstances under which Lazy VFDT is the preferred choice and how it can be integrated into production stream-processing pipelines.

7.2. Summatio

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Wykaz symboli i skrótów

EiTI – Wydział Elektroniki i Technik Informacyjnych
PW – Politechnika Warszawska
WEIRD – ang. *Western, Educated, Industrialized, Rich and Democratic*

Spis rysunków

Spis tabel

Spis załączników

Załącznik 1. Nazwa załącznika 1

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Rysunek 1.1. Obrazek w załączniku.

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Załącznik 2. Nazwa załącznika 2

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Tabela 2.1. Tabela w załączniku.

Kolumna 1	Kolumna 2	Liczba
cell1	cell2	60
	Suma:	123,45

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